SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Short Answer Test

COURSE NAME: COMPUTER VISION **COURSE CODE: DSA0204**

1.Construct a comprehensive and well-organized concept map that accurately identifies all key concepts from literature involved in the pipeline from real-world scene capture to digital image representation, establishes clear and meaningful relationships among intermediate transformations and dependencies, integrates insights from multiple studies, and presents the information in a clear and visually structured manner.

Ans:

Real World Scene

↓

Light Reflection

↓

Camera Lens

↓

Image Sensor (CCD/CMOS)

↓

Analog Signal

↓

Sampling

↓

Quantization

↓

Digital Image

↓

Image Preprocessing

↓

Feature Extraction

↓

Computer Vision Analysis

↓

Output (Recognition / Detection / Classification)

2. Develop a hierarchically structured concept map that identifies major stages of visual data handling in a Computer Vision system, clearly illustrates logical relationships and contributions of each stage toward generating meaningful outputs, incorporates relevant literature connections, and ensures clarity and readability.

Ans:

Image Acquisition

↓

Image Preprocessing

↓

Segmentation

↓

Feature Extraction

↓

Feature Selection

↓

Classification / Recognition

↓

Decision Making

↓

Application Output

3. Create a detailed concept map that accurately identifies concepts related to image resolution and intensity levels, establishes clear cause-effect relationships with image quality and usability, integrates supporting evidence from literature, and presents the concepts in a logically organized and visually clear format.

Ans:

```
Image Resolution
    ├── Pixel Density
    ├── Spatial Resolution
    └── Higher Resolution
            ↓
   Better Image Quality

Intensity Levels
    ├── Bit Depth
    ├── Gray Levels
    └── More Intensity Levels
            ↓
   Better Contrast & Detail

Resolution + Intensity Levels
            ↓
Improved Image Quality
            ↓
Better Computer Vision Performance
```